

TODD A. GONTAREK

GAME ENGINE PROGRAMMER & SOFTWARE ENGINEER

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Portfolio: gontarekt.dev

WORK EXPERIENCE

- Lab Assistant | Graphics Programming with Shaders | Abertay University
September 2026-Present
 - o Supporting students with debugging HLSL shaders & DirectX C++ programs
 - o Demonstrating and explaining graphics concepts to students
- Workshop & Research Assistant | Renewable Parts Ltd.
2022-January 2026
 - o Created macros in Excel to automate data entry, reducing tasks that previously took a full working day to a matter of minutes.

EDUCATION

Abertay University, Dundee, UK

September 2023 – Present

BSc (Hons) Computer Games Technology

- **Professional Project – Grade A:** Programmer in a multi-disciplinary team of 8 developing a game based on a client brief
- **Game Engine Development – Grade A+:** Designed and developed a 3D Game Engine from scratch in C++
- **Graphics Programming with Shaders – Grade A+:** Created custom shaders with shadow mapping and post-processing effects for a DirectX 11 rendering engine
- **Artificial Intelligence – Grade A+:** Created a genetic algorithm & neural network AI from scratch which utilises multi-threading
- **Game Programming and System Architectures – Grade A:** Developed a game with a team of 4 using a PS5 Dev Kit, coded to work cross-platform for PS5 and Windows

SKILLS

Programming Languages: C++, GDScript, C#, HLSL, VBA, HTML, Typescript, Python

Libraries: Entt, Jolt, DirectX 11, SDL3, ImGui, SFML

Game Engines: Godot, Unity

Version Control: Git, Perforce

Miscellaneous: CMake, React, Jira

PERSONAL PROJECTS

- **Galactic Garage** | Godot - Extended from my university client project into a fully networked game
 - o Developed core gameplay mechanics including pickups and a malfunction system
 - o Converted a local multiplayer codebase into a fully networked game
- **Unity Vox Importer**
 - o Voxel importer for MagicaVoxel with configurable voxel output formats
 - o Implemented greedy meshing for efficient voxel rendering

UNIVERSITY PROJECTS

- **Nomad Game Engine** | C++, CMake
 - o Fully Serialized Level Editor, utilizing Entt C++ Reflection Library to dynamically generate UI and dynamically serialize objects without needing a definition of the class
 - o Abstraction Layers, with managers for Input, Audio, Assets, Physics, Rendering, Scenes & Windows
 - o Entity Component System to decouple data from functionality
- **Neuroevolution Snake AI** | Unity, C#
 - o Created a genetic algorithm and neural network from scratch
 - o AI achieved maximum score possible
 - o Optimized the program to train using multi-threading
- **Shader Graphics Project** | C++, DirectX 11
 - o Implemented a dynamic post-processing stack, point light shadow mapping, distance-based terrain tessellation and an object-oriented scene hierarchy

GAME JAMS

Breathing Space	Godot	Week-long 'Finite Space' AGDS 2024 Fresher Game Jam	Team of 2
Sincantation	Unity	Week-long 'Two to One' AGDS 2024 February game jam	Team of 7
Speed Lich	Godot	48 hour 'No End' 2024 Halloween AGDS Game Jam	Team of 4
Born to Krill	Unity	48 hour 'Bubble' Global Game Jam 2025	Team of 5